INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 11

1. Experiment Title:

Other Kinds of Joins

2. Aim:

Use LEFT JOIN, CROSS JOIN, and NATURAL JOIN with team and player tables.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

LEFT JOIN returns all rows from the left table and matching rows from the right table; unmatched right-side columns are NULL. CROSS JOIN returns every possible pair of rows. NATURAL JOIN automatically joins columns with the same name in both tables.

5. Algorithm / Procedure:

1. Create team and player tables and insert the supplied records.
2. Use LEFT JOIN with team on the left so teams without players remain in the output.
3. Use CROSS JOIN to produce every team/player pair.
4. Use NATURAL JOIN to match rows through the common team_id column.

6. Program / Queries / Commands:

```
CREATE TABLE team (
    team_id INT,
    team_name VARCHAR(20)
);

INSERT INTO team (team_id, team_name)
VALUES (1, 'Eagles'), (2, 'Tigers'), (3, 'Lions');

CREATE TABLE player (
    player_name VARCHAR(30),
    team_id INT,
    age INT
);

INSERT INTO player (player_name, team_id, age)
VALUES ('Asha', 1, 24), ('Meera', 1, 22), ('Arjun', 2, 27);

-- Task 1
SELECT t.team_name, p.player_name
FROM team AS t
LEFT JOIN player AS p
    ON t.team_id = p.team_id
ORDER BY t.team_name, p.player_name;

-- Task 2
SELECT t.team_name, p.player_name
FROM team AS t
CROSS JOIN player AS p
ORDER BY t.team_name, p.player_name;

-- Task 3
```

```
SELECT p.player_name, t.team_name
FROM player AS p
NATURAL JOIN team AS t
ORDER BY p.player_name;
```

7. Input:

Teams: Eagles/1, Tigers/2, Lions/3. Players: Asha/1, Meera/1, Arjun/2.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: team_name	player_name
Eagles	Asha
Eagles	Meera
Lions	NULL
Tigers	Arjun
Task 2: team_name	player_name
Eagles	Arjun
Eagles	Asha
Eagles	Meera
Lions	Arjun
Lions	Asha
Lions	Meera
Tigers	Arjun
Tigers	Asha
Tigers	Meera
Task 3: player_name	team_name
Arjun	Tigers
Asha	Eagles
Meera	Eagles

9. Result:

The SQL statements for Experiment 11 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is the difference between LEFT JOIN and INNER JOIN?
2. What is a CROSS JOIN?
3. What does NATURAL JOIN use to match tables?
4. Why does the Lions row show NULL for player_name?